Name: Key

Skills Quiz

1. a) In the following matrix-vector multiplication over the real numbers, find the values of a and b.

$$\begin{pmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \\ 4 & 0 & b \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ a \\ -5 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \\ 4 & 0 & b \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 11 \\ 4+3b \end{pmatrix}$$

So $\boxed{a = 11}$

and $4+3b=-5$, $\boxed{b=-3}$

b) Give an example of a matrix that represents a linear transformation using the standard basis such that its domain is $\mathbb{R}^5$ and its codomain is $\mathbb{R}^2$, or explain why this is not possible.

Any 2×5 real matrix is a correct answer

eg. $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 0 & 0 & 0 & 10 \end{pmatrix}$

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 0 & 0 & 0 & 10 \end{pmatrix} \begin{pmatrix} - \\ - \\ - \\ - \\ - \end{pmatrix} = \begin{pmatrix} - \\ - \end{pmatrix}$$

domain is $\mathbb{R}^5$

codomain is $\mathbb{R}^2$

Name: Key

Skills Quiz

2. a) For each of the following matrix operations, calculate the result if the operation is defined. Otherwise, state that the operation is undefined.

$$A = \begin{pmatrix} 2 & 0 & 3 \\ 0 & 1 & -1 \end{pmatrix} \quad B = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} \quad C = \begin{pmatrix} 2 & 0 & -6 \\ 0 & 1 & 2 \end{pmatrix} \quad D = \begin{pmatrix} 0 & 0 \\ 3 & -2 \\ 0 & 1 \end{pmatrix}$$

$A + C$, AB, DC, BA

$$B+A = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} + \begin{pmatrix} 2 & 0 & 3 \\ 0 & 1 & -1 \end{pmatrix} \leftarrow \text{undefined}$$

$$CD = \begin{pmatrix} 2 & 0 & -6 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 3 & -2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -6 \\ 3 & 0 \end{pmatrix}$$

$$DB = \begin{pmatrix} 0 & 0 \\ 3 & -2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 4 & -2 \\ 1 & 1 \end{pmatrix}$$

$$AB = \begin{pmatrix} 2 & 0 & 3 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} \leftarrow \text{undefined}$$

b) Give an example of two matrices whose product is the matrix M; or explain why this is not possible.

$$M = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix}$$

There are many possible answers. For instance,

$$\begin{pmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} = M$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1/2 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ -1 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$